PERFORMANCE AND BIOINFORMATICS

SSW 565 FALL 2019

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WHAT IS PERFORMANCE?

HOW DOES PERFORMANCE INFORM THE DESIGN OF BIOINFORMATICS SOFTWARE?

CONTENTS

- What Constitutes Performance?
- Tactics
- Why is architecting for performance important?
- Architecting for Performance
- Performance and Bioinformatics
- What is ECM?
- Performance Testing ECM
- Conclusion
- Works Cited

WHAT CONSTITUTES PERFORMANCE

Performance is in part defined by the end-user's perception. It is also defined by quantifiable measures of specific metrics such as latency and resource utilization.

TACTICS

- Architecting performance is achieved using two tactics.
  - Controlling resource demand
  - Managing resource availability

TACTICS:

CONTROLLING RESOURCE DEMAND

- Adjusting data collection rate
- Queueing events
- Binding resource response time

TACTICS:

MANAGING RESOURCE AVAILABILITY

- Parallel utilization of resources
- Redundancy in copies of computations and data
- Limit event arrivals
- Organize resource utilization via schedule

WHY IS ARCHITECTING FOR PERFORMANCE IMPORTANT?

- Scalability
- Throughput
- Stability
- Resource Utilization

ARCHITECTING FOR PERFORMANCE

Performance testing should be integrated into the Software Development Life Cycle

- Identify NFRs
- Build a test environment
- Create scripts for use cases
- Build the test scenario
- Execute testing and analyze results
- Document Results

PERFORMANCE AND BIOINFORMATICS

Bioinformatics applications are inherently high-fidelity data intensive systems. As a result, architecting performance into these systems means managing resources rather than reducing the demand on resources.

WHAT IS ECM?

BASIC OPERATION

- ECM is installed on the Web, File transfer and Application servers.
- ECM interfaces directly with your MS SQL or Oracle database.
- Client components are downloaded from the web browser
- A technology neutral format viewer for raw data is available

(Agilent Technologies ECM 2019)

WHAT IS ECM?

DATA MANAGEMENT AND CHROMATOGRAPHY DATA SYSTEM

OpenLAB ECM
Data Management

Scientific Data Management

- Data Archiving
- Agilent WorkStation Aware!
- Instrument Vendor Neutral
- Time Stamp/Audit Trail
- Report Capture
- Viewers, Filters, Converters
- E-Signatures
- Regulatory Compliance
- TNF Conversion (LTDA)

OpenLAB CDS
Chromatography Data System

Workstations, Client-Server

- ChemStation, EZChrom Elite
- Multi-Vendor
- OpenLAB ECM Intelligent Reporter Regulated and Non-Regulated Modes
- Web-Based Options (ChemLaunch)

(Agilent Technologies ECM 2019)

WHAT IS ECM?

SOFTWARE OVERVIEW

Data Archiving

- Automatic, Manual
- GMP or Non-GMP
- Instrument Vendor Neutral
- Time Stamp / Audit Trail
- Report Capture
- Viewers, Filters, Converters
- eSignatures
- Regulatory Compliance
- Workstation Integration

Document Management

- Excel, Word, PowerPoint, Outlook
- Adobe
- Excel Remediation
- Compliance Documentation
- Image Management (large files)

CDS Integration

- ChemStore
- EZChrom Elite
- CerityP
- Empower
- Dionex
- Galaxy*

Legacy Data Conversion

- TNF Technology
- ANIML
- Viewers
- Reprocessing Options*
- Agilent, HPLAS, Thermo, Waters, Perkin Elmer, Dionex

(Agilent Technologies ECM 2019)

WHAT IS ECM?

KEY SOFTWARE FUNCTIONALITY

- Electronic signatures can be applied to all documents stored in OpenLAB ECM

- Files accessed via the web interface when opened, launch the native application if present on the client

- Smart filters (Attribute Extraction Services) are used to extract key pieces of information from files stored in the repository

- PDF Templates can be configured to extract document specific information as metadata which can be used for searching

(Agilent Technologies ECM 2019)

WHAT IS ECM?

PLATFORM COMPATIBILITY

Overview

ECM Server
- Windows 2016 Server

ECM Client
- Windows 7 SP1 (32-bit or 64-bit)
- Windows 10 (64-bit only)
- Windows 2016 Server

Terminal Server
- Citrix XenApp 7.x
- Microsoft 2016 Terminal Services

Virtualization
- VMWare ESX 6.x

SQL Server Database
- SQL Server 2016
- SQL Server 2016 SP1
- SQL Server 2017

Oracle Database
- Oracle 12c Release 2 Server
- Oracle 12c Release 2 Client (32-bit 12.2.0.1)

NEW IN 12.2

WHAT IS ECM?

PLATFORM COMPATIBILITY – DEVELOPMENT FRAMEWORK

.NET Framework

Required by both ECM Client and Server

- 3.5 Service Pack 1
- 4.5.2 or higher (tested up to 4.7.2)

ECM API

Version 2.2.0.0

- Updated to support CLOB-type User Defined Keys
- Based on .NET 4.5.2 for TLS 1.2 support
- Server-side of API is included with ECM Server base.

ECM API Software Development Kit – Optional purchase/license

Version 2.2.0.0

- Includes updated documentation
- Includes latest ECMAPIWizard

WHAT IS ECM?

HARDWARE COMPONENTS

- Web server
- Application server
- File transfer server
- Database server
- File storage server
- Business Process Manager Server
- Third Party Application Servers (i.e. LIMS)

For small installations, these "virtual servers" can be combined on less than 5 machines. In all cases, it is HIGHLY RECOMMENDED to use an independent database server.

(Agilent Technologies ECM 2019)

WHAT IS ECM?

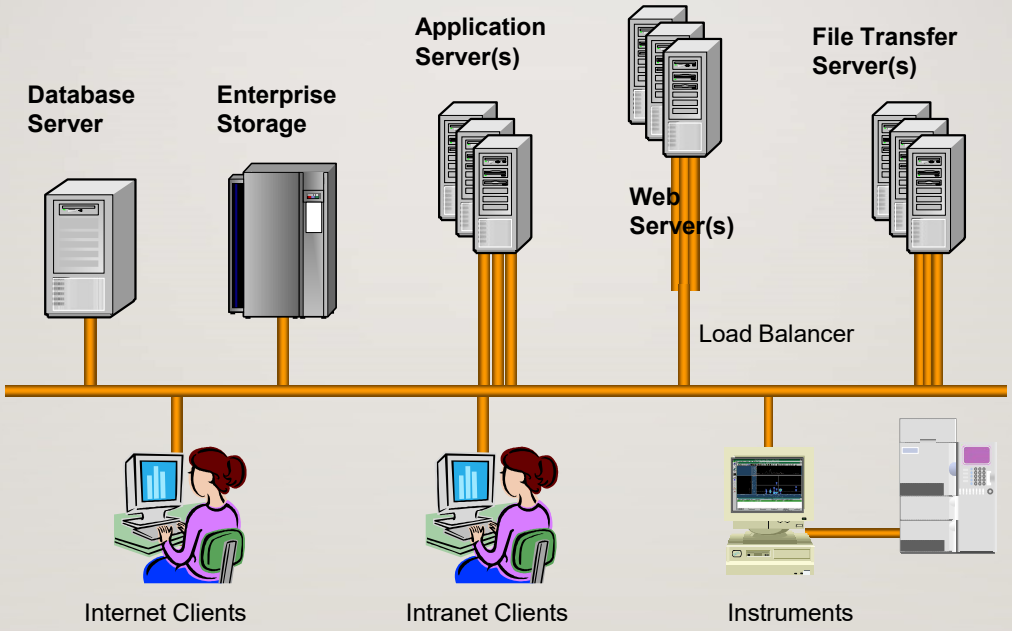
SCALABILITY

- Agilent OL supports multiple web, application, and file transfer servers
- Agilent OL Application and File Transfer servers are load balanced by the Agilent OL software.
- Multiple web servers are supported using NLB.
- Additional servers can be added as the need grows.
- Load balancing for Agilent OL Web Servers is supported using BigIP Load Balancing Switches

(Agilent Technologies ECM 2019)

WHAT IS ECM?

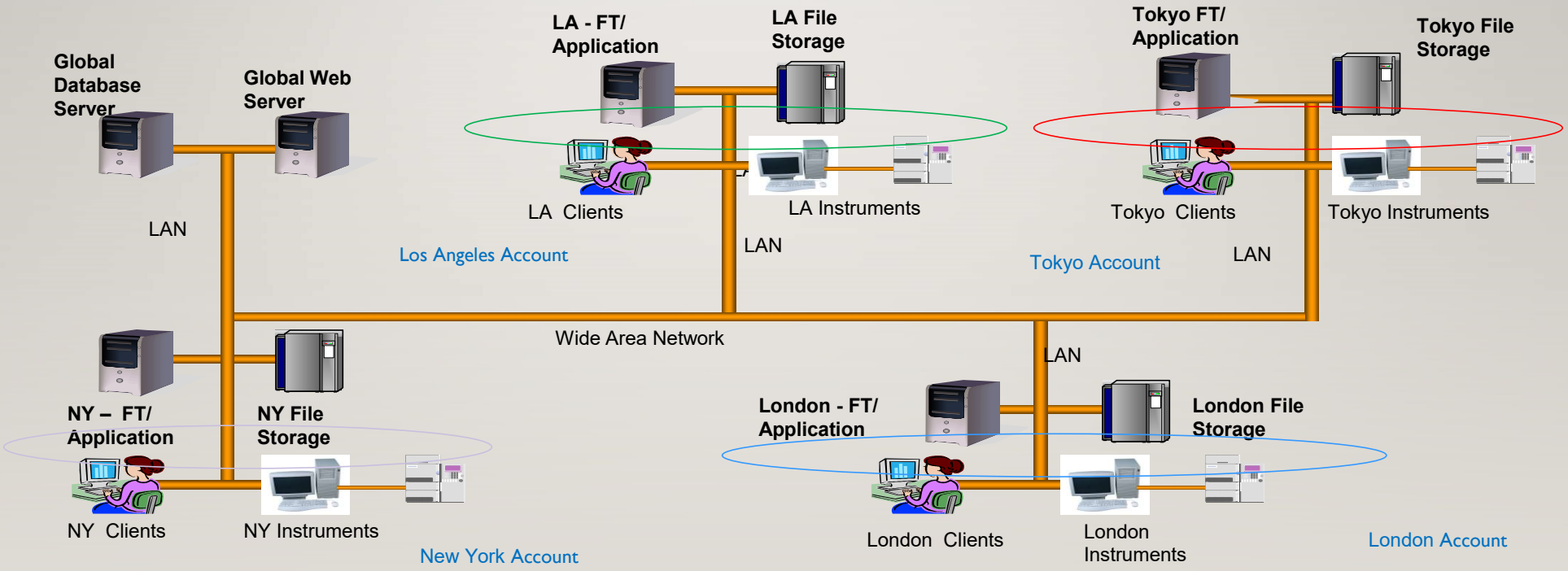
SYSTEM ARCHITECTURE ENTERPRISE IMPLEMENTATION



(Agilent Technologies ECM 2019)

WHAT IS ECM?

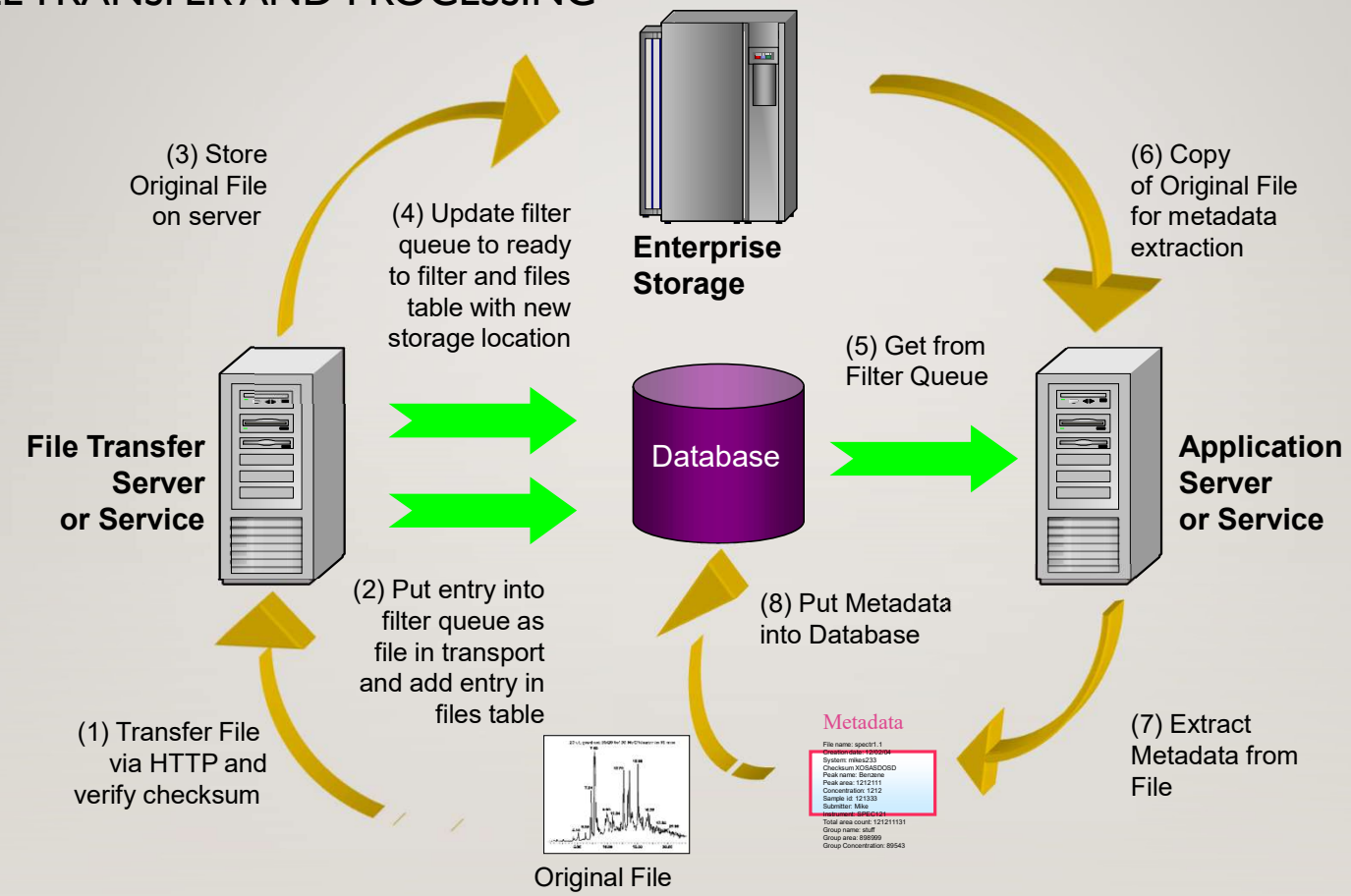
SYSTEM ARCHITECTURE GLOBAL IMPLEMENTATION



(Agilent Technologies ECM 2019)

WHAT IS ECM?

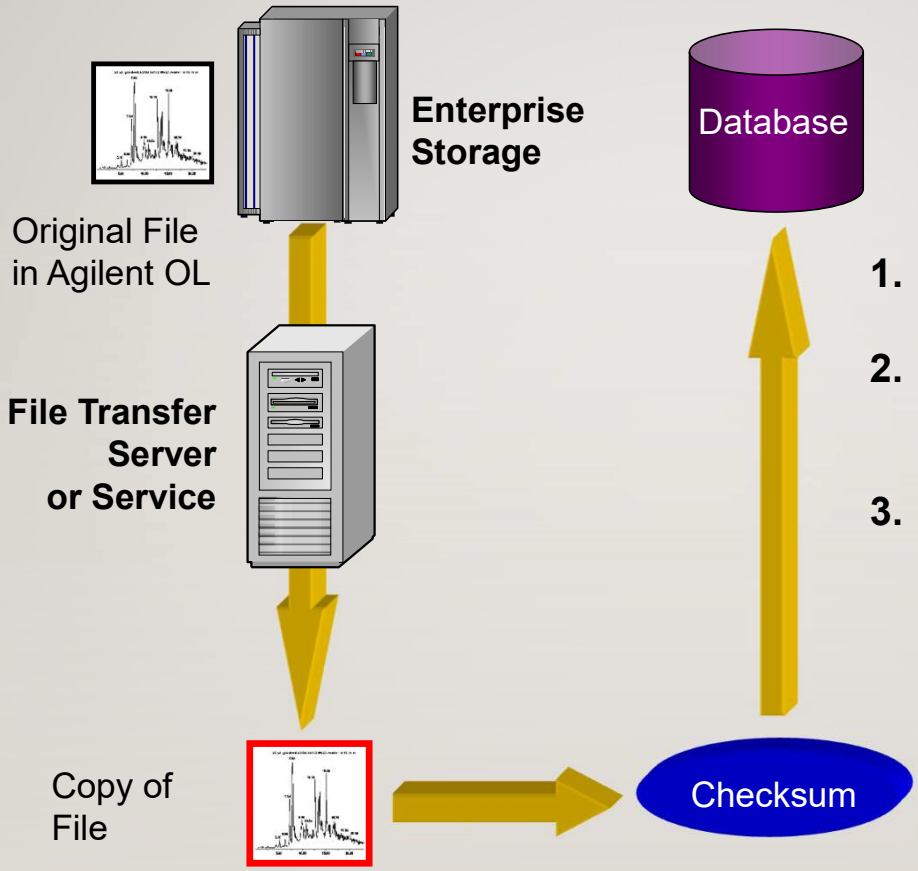
SCALABLE FILE TRANSFER AND PROCESSING



(Agilent Technologies ECM 2019)

WHAT IS ECM?

FILE TRANSFER RELIABILITY

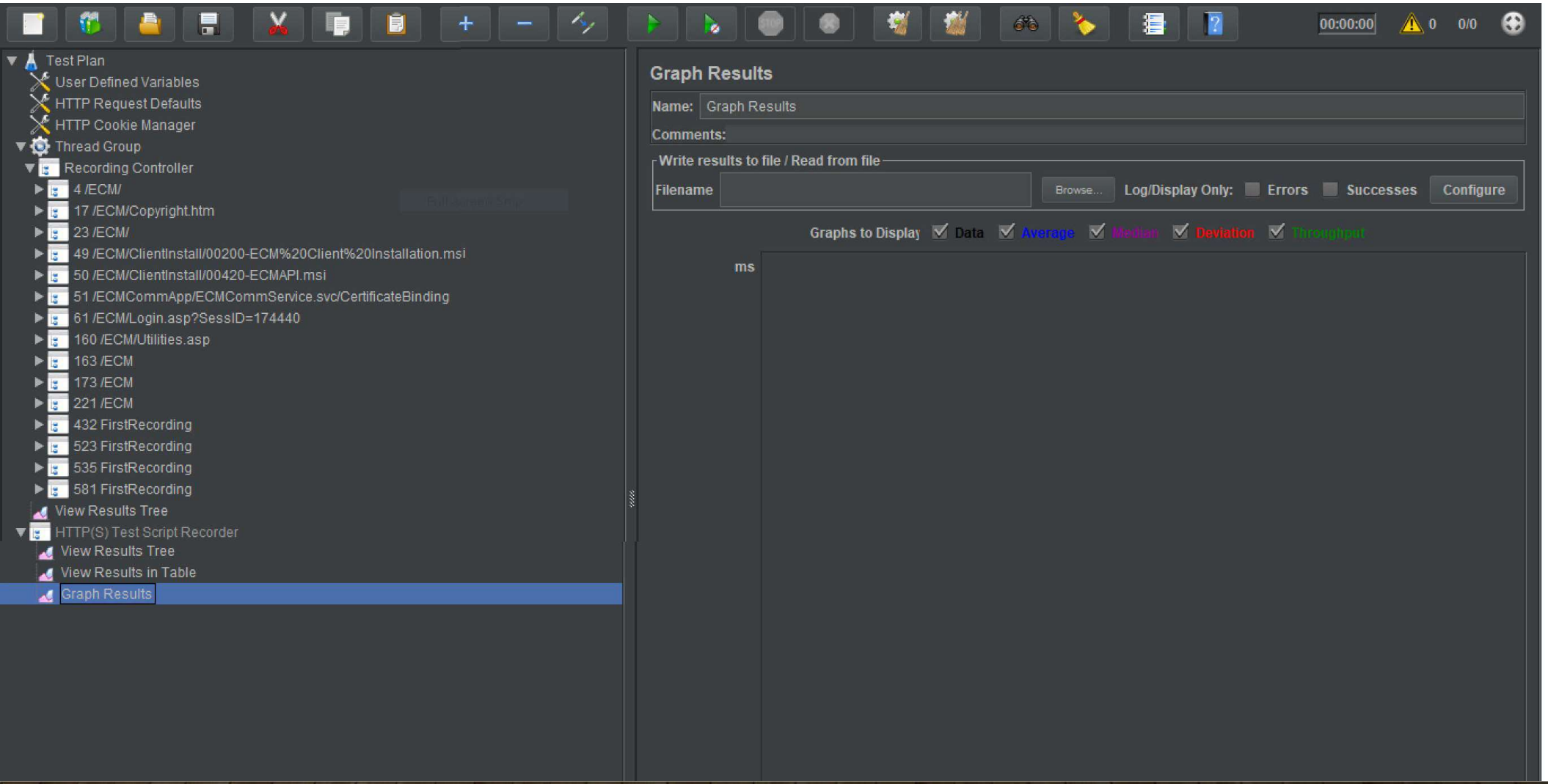


1. **File is copied to client via Transfer Server(s)**
2. **File copy is verified successful thru checksum**
3. **Checksum compared against value stored in database metadata**

(Agilent Technologies ECM 2019)

PERFORMANCE TESTING ECM

Testing the effectiveness of the design choice using the JMeter GUI is very straightforward. The performance testing scenario is transferred to the JMeter platform via creation of the thread group, samplers, logical controllers and listeners. This is accomplished using a user interface and intuitive dialogue boxes. There also exist templates for gaining a better understanding of the Web API you are testing.



JMeter: ECM Recording Controller Test Plan

ECM Login.jmx (C:\Users\Anthem Rukiya\Downloads\apache-jmeter-5.1.1\apache-jmeter-5.1.1\bin\ECM Login.jmx) - Apache JMeter (5.1.1 r1855137)

File Edit Search Run Options Tools Help

00:00:00 0 0/0

- Recording Controller
  - 4 /ECM/
  - 17 /ECM/Copyright.htm
  - 23 /ECM/
  - 49 /ECM/ClientInstall/00200-ECM%20Client%20Installation.msi
  - 50 /ECM/ClientInstall/00420-ECMAPI.msi
  - 51 /ECMCommApp/ECMCommService.svc/CertificateBinding
  - 61 /ECM/Login.asp?SessID=174440
    - 61 /ECM/Login.asp?SessID=174440
      - HTTP Header Manager
    - 62 /ECM/Install.asp
    - 63 /msadc/msadcs.dll/CyberLabRDS.CBO.Initialize
    - 64 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 65 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 66 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 68 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 67 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 72 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 71 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 70 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 69 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 73 /ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
    - 74 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 75 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 76 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 77 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 78 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 79 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 80 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 81 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 82 /ECM/Enterprise.asp
    - 83 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 84 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 86 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 85 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 87 /ECM/components/SSIKESExplorer.lpk
    - 88 /ECM/HeartBeat.asp
    - 89 /ECM
    - 90 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 91 /ECMCommApp/ECMCommService.svc/CertificateBinding
    - 92 /ECMCommApp/ECMCommService.svc/CertificateBinding

HTTP Request

Name: 61 /ECM/Login.asp?SessID=174440

Comments:Detected the start of a redirect chain

Basic | Advanced

Web Server

Protocol [http]: | Server Name or IP: win-43g480f7bpe | Port Number:

HTTP Request

Method: POST | Path: /ECM/Login.asp?SessID=174440 | Content encoding:

☐ Redirect Automatically ☑ Follow Redirects ☑ Use KeepAlive ☐ Use multipart/form-data ☐ Browser-compatible headers

Parameters | Body Data | Files Upload

Send Parameters With the Request:

Name:	Value	URL Encode?	Content-Type	Include Equals?
dfName	admin	☐	text/plain	☑
dfPassword	admin	☐	text/plain	☑
dfAccount	admin	☐	text/plain	☑
dfDomainName	Built-In	☐	text/plain	☑

Detail | Add | Add from Clipboard | Delete | Up | Down

Type here to search

6:17 PM
11/3/2019

JMeter: ECM Recording Controller Test Plan Results

Thread Group
View Results Tree
HTTP(S) Test Script Recorder
View Results Tree
View Results in Table
Graph Results

Comments:

Write results to file / Read from file

Filename recording.xml Browse... Log/Display Only: Errors Successes Configure

Search: Case sensitive Regular exp. Search Reset

Text

clients4.google.com:443
432 FirstRecording
433 FirstRecording
434 FirstRecording
435 FirstRecording
436 FirstRecording
437 FirstRecording
438 FirstRecording
439 FirstRecording
442 FirstRecording
441 FirstRecording
440 FirstRecording
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448 FirstRecording
449 FirstRecording
450 FirstRecording
451 FirstRecording
452 FirstRecording
453 FirstRecording
454 FirstRecording
455 FirstRecording
456 FirstRecording

Scroll automatically?

Sampler result | Request | Response data

Request Body | Request Headers

Find Case sensitive Regular exp.

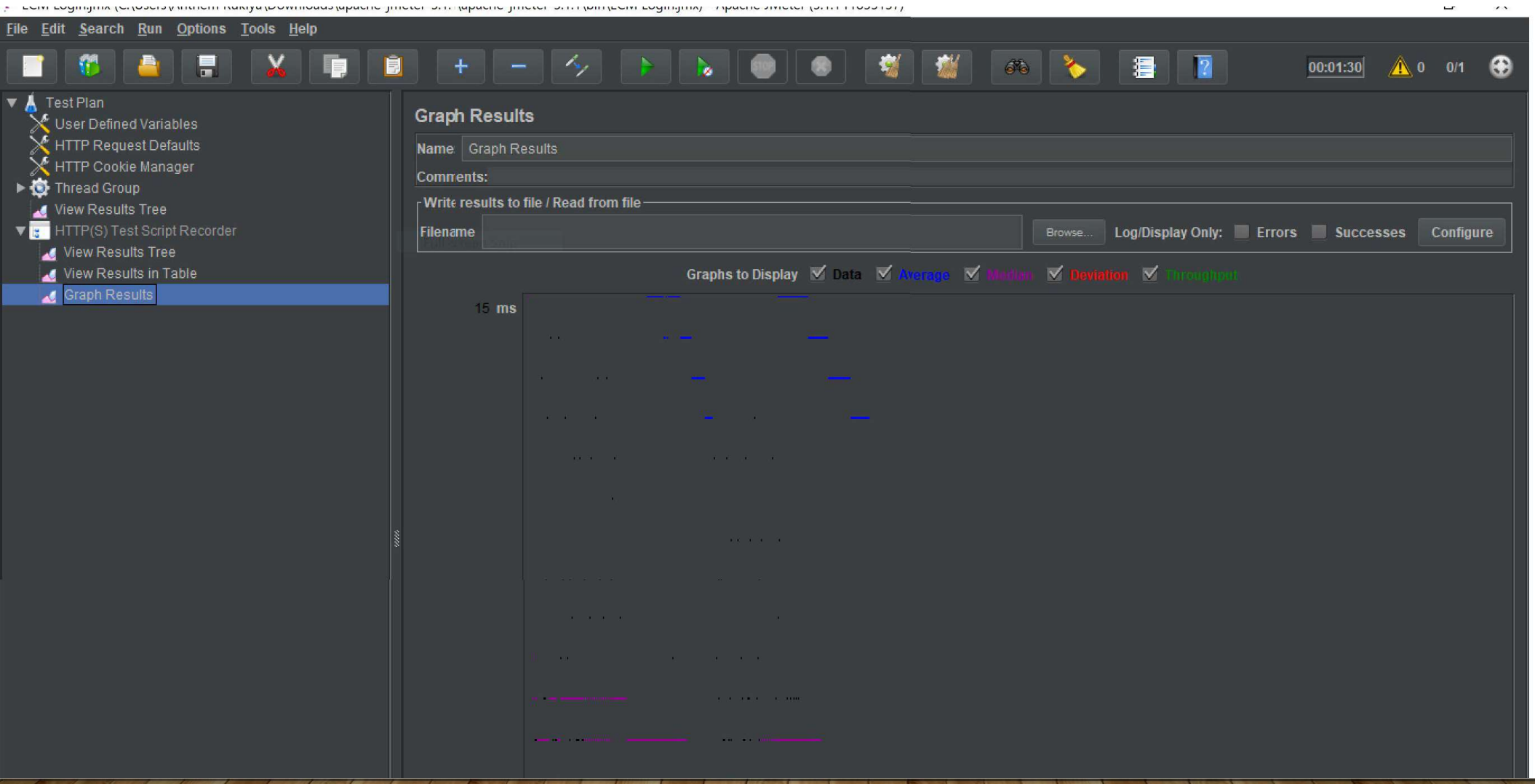
```
POST http://win-43g480f7bpe/ECMCommApp/ECMCommServiceTester.svc/CertificateBinding

POST data:
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope" xmlns:a=
"http://www.w3.org/2005/08/addressing"><s:Header><a:Action s:mustUnderstand="1">http:
//schemas.xmlsoap.org/ws/2005/02/trust/RST/Issue</a:Action><a:MessageID>urn:uuid:f2909b81-d2f7-4282
-8da1-4367ac1b208d</a:MessageID><a:ReplyTo><a:Address>
http://www.w3.org/2005/08/addressing/anonymous</a:Address></a:ReplyTo><a:To s:mustUnderstand="1">
http://win-43g480f7bpe/ECMCommApp/ECMCommServiceTester.svc/CertificateBinding
</a:To></s:Header><s:Body><t:RequestSecurityToken Context="uuid-5da93bbd-eedf-4a03-8765-d949ce63439
f-323" xmlns:t="http://schemas.xmlsoap.org/ws/2005/02/trust"><t:TokenType>
http://schemas.xmlsoap.org/ws/2005/02/sc/sct</t:TokenType><t:RequestType>
http://schemas.xmlsoap.org/ws/2005/02/trust/Issue
</t:RequestType><t:KeySize>256</t:KeySize><t:BinaryExchange ValueType="
http://schemas.xmlsoap.org/ws/2005/02/trust/tlsnego" EncodingType="
http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-soap-message-security-1.0#Base64Binary
">FgMBAFoBAABWAwFdv1ubGJoyxlTw9H96XGsrhlETVJ9mPyL1hAOkxKSRSgAADsAKwAnAFMATADUALwAKAQAAHwAKAAgABgAdA
BcAGAALAAIBAAAjAAAAFwAA/wEAAQA=</t:BinaryExchange></t:RequestSecurityToken></s:Body></s:Envelope>

[no cookies]
```

Raw | HTTP

JMeter: ECM Recording Controller Test Plan Raw Data



JMeter: ECM Recording Controller Test Plan Graph

ECM Login.jmx (C:\Users\Anthem Rukiya\Downloads\apache-jmeter-5.1.1\apache-jmeter-5.1.1\bin\ECM Login.jmx) - Apache JMeter (5.1.1 r1855137)

File Edit Search Run Options Tools Help

00:01:30 0 0/1

- Test Plan
  - User Defined Variables
  - HTTP Request Defaults
  - HTTP Cookie Manager
  - Thread Group
  - View Results Tree
  - HTTP(S) Test Script Recorder
    - View Results Tree
    - View Results in Table
    - Graph Results

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename Browse... Log/Display Only: Errors Successes Configure

Sample #	Start Time	Thread Name	Label	Sample Time(m...	Status	Bytes	Sent Bytes	Latency	Connect Time(...
1	19:00:00.000		clients4.google...	0		794	0	0	0
2	17:58:34.621		432 FirstRecor...	1097		1951	2118	1097	1032
3	17:58:35.721		433 FirstRecor...	15		8347	3264	15	1
4	17:58:35.779		434 FirstRecor...	3		1486	5289	2	1
5	17:58:36.202		435 FirstRecor...	22		2881	1502	22	2
6	17:58:36.233		436 FirstRecor...	5		3368	2831	4	0
7	17:58:36.242		437 FirstRecor...	6		7562	6940	6	1
8	17:58:36.252		438 FirstRecor...	4		3597	4093	4	1
9	17:58:36.260		439 FirstRecor...	4		6775	6965	4	1
10	17:58:36.283		442 FirstRecor...	3		3669	4139	3	1
11	17:58:36.275		441 FirstRecor...	4		6689	7491	4	1
12	17:58:36.268		440 FirstRecor...	4		6850	7524	4	1
13	17:58:36.298		444 FirstRecor...	4		3359	4418	3	1
14	17:58:36.292		443 FirstRecor...	3		3375	4170	3	1
15	17:58:36.466		445 FirstRecor...	13		2881	1490	13	1
16	17:58:36.488		446 FirstRecor...	5		3368	2819	5	1
17	17:58:36.498		447 FirstRecor...	5		7562	6928	5	1
18	17:58:36.506		448 FirstRecor...	4		3591	4081	4	1
19	17:58:36.514		449 FirstRecor...	8		6769	6947	8	1
20	17:58:36.527		450 FirstRecor...	12		6809	7619	11	0
21	17:58:36.545		451 FirstRecor...	21		9483	7649	21	1
22	17:58:36.596		452 FirstRecor...	27		13924	7649	26	1
23	17:58:36.662		453 FirstRecor...	14		7300	5474	14	1
24	17:58:36.687		454 FirstRecor...	4		6683	7473	4	0
25	17:58:36.695		455 FirstRecor...	5		3669	4127	5	1
26	17:58:36.703		456 FirstRecor...	4		3375	4158	4	1
27	17:58:36.711		457 FirstRecor...	4		3359	4406	4	0
28	17:58:36.868		458 FirstRecor...	3		433	3590	3	1
29	17:58:37.074		459 FirstRecor...	2		302	118	0	0
30	17:58:37.136		460 FirstRecor...	14		2881	1490	13	1
31	17:58:37.158		461 FirstRecor...	6		3368	2819	5	1
32	17:58:37.168		462 FirstRecor...	5		7562	6928	5	1
33	17:58:37.177		463 FirstRecor...	3		3591	4081	3	0
34	17:58:37.184		464 FirstRecor...	8		6769	6947	7	1
35	17:58:37.355		465 FirstRecor...	2		302	118	0	1

Scroll automatically? Child samples? No of Samples 307 Latest Sample 0 Average 12 Deviation 85

Type here to search

6:02 PM
11/3/2019

JMeter: ECM Recording Controller Test Plan Results Table

ECM Login.jmx (C:\Users\Anthem Rukiya\Downloads\apache-jmeter-5.1.1\apache-jmeter-5.1.1\bin\ECM Login.jmx) - Apache JMeter (5.1.1 r1855137)

File Edit Search Run Options Tools Help

00:00:09 0 0/6

- ECM Test
  - Thread Group
    - Once Only Controller
      - ECM Login
        - ECM Login Results
        - View Results Tree

HTTP Request

Name: ECM Login

Comments:

Basic | Advanced

Web Server

Protocol [http]: Server Name or IP: http://win-43g480f7bpe/ECM/ Port Number:

HTTP Request

Method: POST Path: Content encoding:

☐ Redirect Automatically ☑ Follow Redirects ☑ Use KeepAlive ☐ Use multipart/form-data ☐ Browser-compatible headers

Parameters | Body Data | Files Upload

Send Parameters With the Request:

Name:	Value	URL Encode?	Content-Type	Include Equals?
username:	admin	☐	text/plain	☑
password:	admin	☐	text/plain	☑
account:	admin	☐	text/plain	☑
domain:	Built-In	☐	text/plain	☑

JMeter: ECM Login Test Plan

251 /ECM/Login.asp
HTTP Header Manager
View Results in Table
Graph Results
View Results Tree

Comments:

Write results to file / Read from file

Filename | Browse... | Log/Display Only: Errors Successes | Configure

Search: | Case sensitive | Regular exp. | Search | Reset

Text

251 /ECM/Login.asp
251 /ECM/Login.asp-0
251 /ECM/Login.asp-1
251 /ECM/Login.asp-2
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
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251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp
251 /ECM/Login.asp

Sampler result | Request | Response data

Request Body | Request Headers

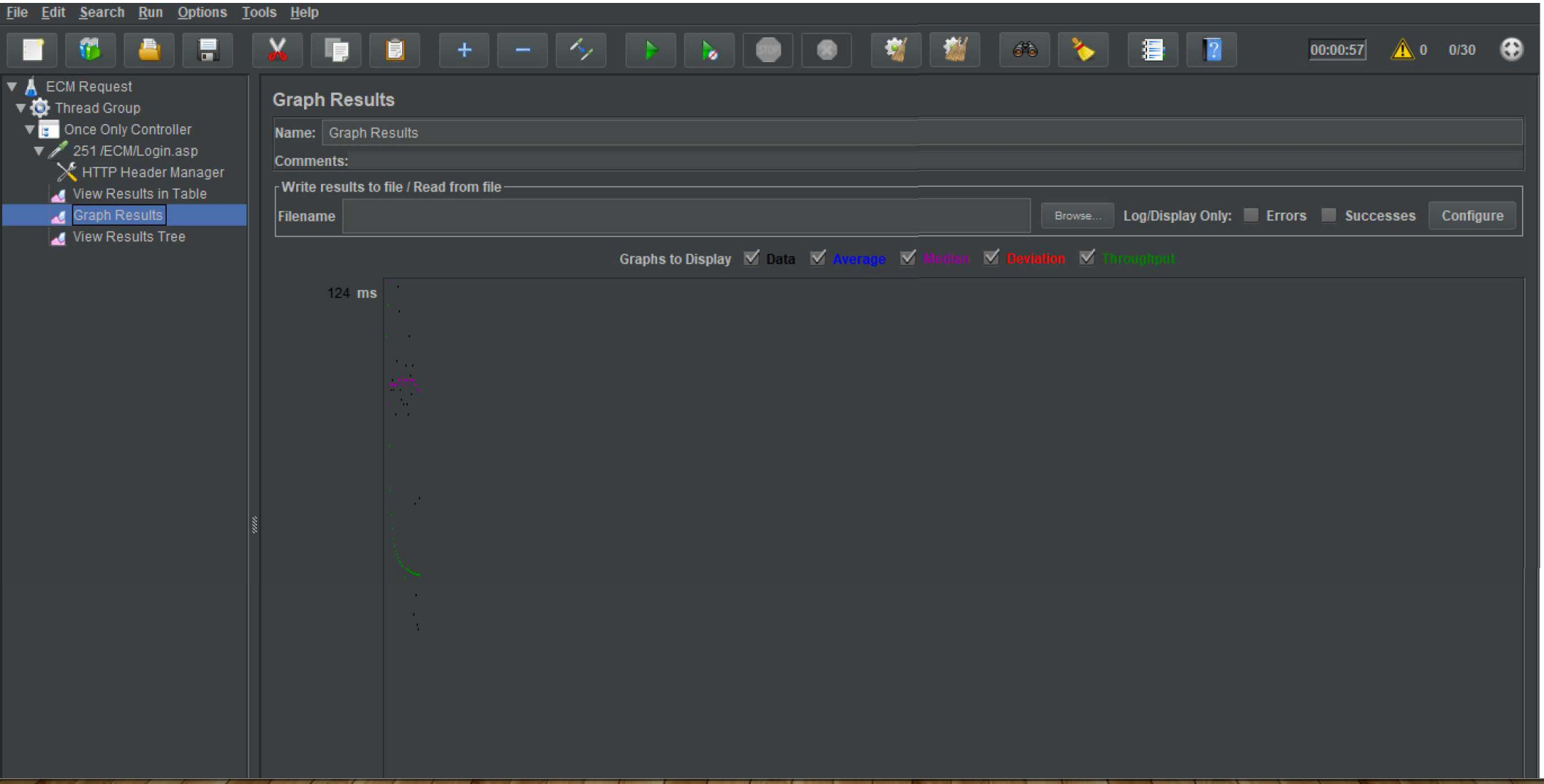
Find | Case sensitive | Regular exp.

```
POST http://win-43g480f7bpe/ECM/Login.asp

POST data:
dfName=admin&dfPassword=admin&dfAccount=admin&dfDomainName=Built-In

[no cookies]
```

JMeter: ECM Login Test Plan Raw Data



JMeter: ECM Login Test Plan Graph

ECM Login 2.jmx (C:\Users\Anthem Rukiya\Downloads\apache-jmeter-5.1.1\apache-jmeter-5.1.1\bin\ECM Login 2.jmx) - Apache JMeter (5.1.1 r1855137)

File Edit Search Run Options Tools Help

00:00:57 0 0/30

- ECM Request
  - Thread Group
    - Once Only Controller
      - 251 /ECM/Login.asp
        - HTTP Header Manager
        - View Results in Table
        - Graph Results
        - View Results Tree

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename | Browse... | Log/Display Only: Errors Successes | Configure

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	18:24:40.224	Thread Group 1-1	251 /ECM/Login.a...	1097		7322	1828	1052	998
2	18:24:42.204	Thread Group 1-2	251 /ECM/Login.a...	99		7322	1828	52	1
3	18:24:44.204	Thread Group 1-3	251 /ECM/Login.a...	124		7322	1828	79	2
4	18:24:46.203	Thread Group 1-4	251 /ECM/Login.a...	103		7322	1828	58	3
5	18:24:48.205	Thread Group 1-5	251 /ECM/Login.a...	102		7322	1828	58	2
6	18:24:50.204	Thread Group 1-6	251 /ECM/Login.a...	104		7322	1828	60	3
7	18:24:52.210	Thread Group 1-7	251 /ECM/Login.a...	102		7322	1828	55	2
8	18:24:54.204	Thread Group 1-8	251 /ECM/Login.a...	128		7322	1828	82	2
9	18:24:56.204	Thread Group 1-9	251 /ECM/Login.a...	97		7322	1828	54	2
10	18:24:58.202	Thread Group 1-10	251 /ECM/Login.a...	108		7322	1828	62	1
11	18:25:00.204	Thread Group 1-11	251 /ECM/Login.a...	123		7322	1828	79	2
12	18:25:02.206	Thread Group 1-12	251 /ECM/Login.a...	118		7322	1828	73	2
13	18:25:04.203	Thread Group 1-13	251 /ECM/Login.a...	102		7322	1828	55	2
14	18:25:06.203	Thread Group 1-14	251 /ECM/Login.a...	100		7322	1828	53	2
15	18:25:08.204	Thread Group 1-15	251 /ECM/Login.a...	104		7322	1828	59	2
16	18:25:10.204	Thread Group 1-16	251 /ECM/Login.a...	99		7322	1828	57	2
17	18:25:12.203	Thread Group 1-17	251 /ECM/Login.a...	1190		7322	1828	1137	1086
18	18:25:14.204	Thread Group 1-18	251 /ECM/Login.a...	107		7322	1828	60	2
19	18:25:16.219	Thread Group 1-19	251 /ECM/Login.a...	99		7322	1828	55	2
20	18:25:18.219	Thread Group 1-20	251 /ECM/Login.a...	97		7322	1828	55	2
21	18:25:20.220	Thread Group 1-21	251 /ECM/Login.a...	113		7322	1828	72	2
22	18:25:22.204	Thread Group 1-22	251 /ECM/Login.a...	105		7322	1828	59	2
23	18:25:24.205	Thread Group 1-23	251 /ECM/Login.a...	101		7322	1828	57	2
24	18:25:26.205	Thread Group 1-24	251 /ECM/Login.a...	107		7383	1828	65	2
25	18:25:28.205	Thread Group 1-25	251 /ECM/Login.a...	56		4830	658	56	2
26	18:25:30.205	Thread Group 1-26	251 /ECM/Login.a...	79		4830	658	79	2
27	18:25:32.204	Thread Group 1-27	251 /ECM/Login.a...	60		4830	658	60	2
28	18:25:34.204	Thread Group 1-28	251 /ECM/Login.a...	54		4830	658	54	2
29	18:25:36.203	Thread Group 1-29	251 /ECM/Login.a...	53		4830	658	53	1
30	18:25:38.203	Thread Group 1-30	251 /ECM/Login.a...	80		4830	658	80	2

Scroll automatically? Child samples? No of Samples 30 Latest Sample 80 Average 167 Deviation 261

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JMeter: ECM Login Test Plan Results Table

PERFORMANCE TESTING ECM:

TEST PLAN

- Thread Groups Utilized (3 Scenarios)
  - 1 Thread/ 1 Second
  - 6 Threads / 60 Seconds
  - 30 Threads/60 Seconds
- Once Only Logical Controller
  - HTTP Request – "ECM Login"
  - Server/IP
    - http://win-43g480f7bpe/ECM/
  - Post
    - User name
    - Password
    - Account
    - Domain
- Listener
  - Graph Results – "ECM Login Results"
  - View Results Tree

PERFORMANCE TESTING ECM:

TEST RESULTS

A sample population of thirty yields an average service time of 167 milliseconds. The throughput was recorded as 0.52 transactions per second with the system performing under a moderate load. For this test scenario, server utilization is calculated to be 0.087. The average response time was a recorded 2.82 seconds. Thus the average queue length shall then be calculated to be 1.4664 events. If the arrival rate for the scenario is roughly 0.024 events per second based on the calculated average queue length, then the calculated traffic intensity is 0.0041. For a moderate load, this measure of traffic intensity reveals that the system is not at all saturated.

PERFORMANCE TESTING ECM:

HOW CAN ECM BE IMPROVED?

Supporting all in one ECM servers with domain-independent configurations ready to deploy as a super-light topology for the cloud, is the would be the best next step to optimize for performance. Providing a domain-independent solution that is not reliant on ASP.Net and IIS for Windows Authentication will allow customers to take the secondary step of moving their entire bioinformatics system to a cloud-based topology.

CONCLUSION

Every attempt to make a system more efficient may reveal or create an unknown issue that diminishes the overall optimization effort. Andre Bondi provides this practical advice for performance engineering efforts, designing architecture on the basis of performance is an effort which must be made with "vigilance and skepticism". Performance engineering isn't about reshaping an entire system, so much as it is about improving the overall efficiency of a system while making the smallest possible impact upon it.

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